

LiveC Requirements Document

Rough draft 6-4-2025

[Note: Sections 5, 6, 7, and 10 include material specifically developed for this project; the rest are from a template that Dana provided.]

1. Project Overview

Brief description of the project/product main goals.

2. Overview

Provide an overview of the sections and contents of the document. Alternatively, a table of contents here instead.

3. Business Requirements

- i. **Background** Summarize the rationale and context for the new product or for changes to be made to an existing one.
- ii. **Business Opportunity** Describe the business problem that is being solved or the process being improved. Describe the needs of typical customers or of the target market.
- iii. **Business Objectives** Summarize the important business benefits the product will provide in a quantitative and measurable way. Organizations generally undertake a project to solve a problem or exploit an opportunity. Business objectives define ways to measure achievement of business goals.
- iv. **Success Metrics** Indicators that stakeholders will use to define and measure success on the project. What are the factors that will have the greatest impact on achieving success (i.e., could include aspects within and external to the organization).
- v. **Product Vision Statement** Write a concise vision statement that summarizes the long-term purpose and intent of the product. A simple diagram that shows the major components of the overall system, subsystem interconnections, and external interfaces can be helpful but not required.

4. Scope and Limitations

- i. **Major Features** Summarize the major features the product contains or the significant functions that it performs or lets the user perform. Details will be provided in Section 5, so only a high level summary is needed here.
- ii. **Project Scope** Provide a short description of the software being specified and its purpose, including relevant benefits, objectives, and goals. Relate the software to corporate goals or business strategies.
- iii. **Limitations and Exclusions** List any product capabilities or characteristics that a stakeholder might expect but that are not planned for inclusion in the product or in a specific release. List items that were cut from scope, so the scope decision is not forgotten.

5. Context Description

i. User Classes and Characteristics

Community Member: Provides feedback in the form of a Proposed Change, on an ongoing basis

ACM Ed Board: Recruits 1 Editor-in-Chief for each discipline

Editor-in-Chief: Recruit X Associate Editors for each discipline, forwards updated curriculum to ACM Staff for posting

Associate Editor: Provides notification of the Proposed Change being received and that it is being reviewed, Reviews contributions and/or Recruits external reviewers, provides notification of proposed change being accepted for inclusion in the next version

Reviewer: recruited by the Associate Editor, provides a recommendation of inclusion in the next version to the Associate Editor

Editorial Board: Editor in Chief + all the Associate Editors, review feedback accumulated over the entire cycle

ACM Support Staff

Ii. Live Curricula Update Process

It is expected to have an annual update cycle.

Each cycle will be run by Editors-in-Chief, one for each discipline; by associate editors nominated by the Editors-in-Chief, with the assistance of

external reviewers in cases where the associate editor thinks they cannot judge the contribution.

The ACM Ed Board or the ACM EAC will nominate the Editors-in-Chief. They, in turn, will nominate the associate editors.

The frequency for updating once a year may not be mandatory. It is possible that each discipline will select a frequency that best suits its needs. This is one of the questions that needs to be discussed by the entire group.

The same goes for the number of associate editors recruited.

Stage	Description	Comments	Actors	Frequency ¹
1. Launch of the Cycle				
1a	Place updated curricula on the site for each of the disciplines		ACM Support Staff	Once every Y years
1b	Dissemination - First time publicizing to the community	Post to SIG listservs, ACM MemberNet and similar announcements	ACM Ed Board? ACM EAC	Once every Y years
1c	Recruit 1 Editor-in-Chief for each discipline		ACM Ed Board	Once every Y years
1d	Recruit X Associate Editors for each discipline		Editor-in-Chief	Once every Y years
1e	Recruit external Reviewers		Associate Editors	As needed
2. Community Contributions				
2a	Community provides feedback		Community members	Daily/ongoing
2b	Acknowledgment	Public notification of the feedback	Associate Editors	Once every Z months

		that has been received and that it is being reviewed		
3. Review of Contributions				
3a	Review contributions		Associate Editors	Once every Z months
3b	Decision/Recommendation	Public notification of exclusion or consideration for inclusion, not necessarily inclusion	Associate Editors	Once every Z months
4. Inclusion of Contributions				
4a	Review of feedback accumulated over the entire cycle	Cross-KA issues to be considered for inclusion of feedback	Editorial Board consisting of Editor-in-Chief and all the Associate Editors	Once every Y years
4b	Decision disseminated	Public notification of the feedback accepted for inclusion in the next version	Associate Editor	Once every Y years
4c	Decision incorporated	Next version of the curriculum is updated as per the feedback	Associate Editor	Once every Y years
4d	Contributors acknowledged in the next version of the curriculum	In a Contributors table	Associate Editor	Once every Y years
4e	Updated curriculum forwarded to ACM Staff for posting (Step 1a)		Editor-In-Chief	Once every Y years

¹ Each discipline will determine its own process pacing.

ii. **Operating Environment**

Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.

iii. **Design and Implementation Constraints**

Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).

iv. **Assumptions and Dependencies**

List any assumed factors (as opposed to known facts) that could affect the requirements stated. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project.

v. **Glossary of Terms**

Define all the terms, including acronyms and abbreviations.

vi. **References**

6. System Features

A number of system features are listed below together with more detailed user stories. Note that User story numbers are used within the activity diagrams in Section 10.

System Feature 1: LiveC shall allow a community member to select a specific discipline and provide a recommendation in the form of a proposed change to a curriculum.

Associated user stories in Table below:

User story #	Process Code (references the process model)	User story	Acceptance criteria / Comments / Questions
CM		Community Member	
CM1	2a	As a community member, I want to register as a user of the system so that I can use the full set of system services.	Who are eligible to register? E.g., only members of the professional/academic societies?
CM2	2a	As a community member, I want to select a specific curriculum recommendation (CS 2023, IS 2020, etc.) for review so that I can propose changes.	
CM3	2a	As a community member, I want to access a specific section of the curriculum using multiple mechanisms (sequential reading, text search, table of contents, index, etc.) so that I can easily access sections that I am interested in reviewing.	
CM4	2a	As a community member, I want to be able to provide feedback at multiple levels	The hierarchies vary depending on the discipline. Most of them are either

		of hierarchy, including proposing a) new elements, b) changes to the sequence of the elements, c) removal of elements, and d) detailed textual changes so that I can easily articulate the type of change I am proposing	competency (e.g., competency realm, competency area, competency) or knowledge (e.g., knowledge area, knowledge unit, topics+learning outcomes) hierarchies
CM5	N/A	As a community member, I want to download the latest version of a specific curriculum recommendation so that I can use it to support my work.	

System Feature 2: LiveC shall process the proposed change by

- a. Initiating a public notification that the community member's proposed change was received and is being considered
- b. Allow an Associate Editor to triage the proposed change and decide: desk reject, AE review or external reviewer review.
- c. Allow an Associate Editor to Invite external reviewers (if decided at b.)
- d. Collect assessment from reviewers and make it available to the Associate Editor
- e. Allow the Associate Editor to submit recommendation (decision) for inclusion/exclusion in next version of curriculum to the Editor in Chief
- f. Send notification to the Community member who initiated the Proposed change as well as the Community as to the decision at point e.

Associated user stories in Table below:

AE / R		Associate Editor and Reviewer	
AE1	2b	As an AE, I want to send a public notification	

		regarding the feedback that has been provided and personal acknowledgement to the community member	
AE2	3a	As an AE, I want to triage the proposed changes that have been submitted regarding my area of responsibility since the previous review so that I can specify the actions appropriate for each proposed change (such as desk reject, AE review, reviewer review).	
R1	3a	As a reviewer, I want to read and respond to an AE's invitation to review a proposed change.	
R2	3a	As a reviewer, I want to evaluate the merits of each proposed change in my queue so that I can provide my feedback regarding the proposed changes.	The reviewer should be able to move easily between all proposals in the queue and the history of changes of the target segment.
R3	3a	As a reviewer, I want to enter my recommendation regarding a proposed change and its justification and submit them to the AE so that the review elements are collected systematically.	
AE3	3a	As an AE, I want to evaluate the merits of each proposed change in my queue so that I can	Essentially the same as R2

		document my feedback regarding the proposed changes.	
AE4	3b	As an AE, I want to enter my recommendation regarding a change proposal and its justification so that it is available for overall review	Essentially the same as R3
AE5	3b	As an AE, I want to review all feedback (mine and reviewers) provided regarding proposed changes, finalize and submit my final recommendation to the EIC, and provide appropriate public feedback.	This is particularly important for those proposals that have
AE6	As needed	As an AE, I want to review the entire proposal and decision history regarding my area of responsibility so that I can consider prior decisions and current proposals at various stages of the decision process.	Available at all stages of the process; not a separate process step
AE7	4c	As an AE, I want to update the curriculum within my area of responsibility so that the approved changes can be integrated to the curriculum recommendation.	
AE8	4a	As an AE, I want to respond to a revision request by the EIC with changes to the text and/or a justification.	

AE9	3a	As an AE, I want to document the justifications for desk rejected proposed change.	
EIC		Editor-in-Chief	
EIC1	4a	As the EIC, I want to evaluate the recommendations by the AEs and make my individual decisions regarding each proposal in preparation for the Editorial Board decision process so that I am ready to lead the discussion.	
EIC2	4a	As the EIC, I want to review all changes made to a section of my curriculum recommendation by an AE and either approve the full recommendation or send change request(s) to the AE	There can be multiple rounds of reviews
EIC3	4a	As the EIC, I want to present the reviewed recommendations to the Editorial Board for their final decision and document the Board's decisions in the system so that the decisions are immediately captured.	
EIC4	As needed	As the EIC, I want to review the entire proposal and decision history for my computing discipline so that I can consider prior decisions and current	Available at all stages of the process

		proposals at various stages of the decision process.	
EIC5		As the EIC, I want to approve a specific collection of curriculum elements as the next official version.	Society approvals?
EIC6	4b	As the EIC, I want to use the LiveC site to notify the public regarding the approved changes to the curriculum so that uses of the curriculum recommendation stay up-to-date regarding the developments	
EIC7		As the EIC, I want to acknowledge received proposed changes from an AE	
AS		ACM Staff	How about other organizations?
AS1	4e	As the relevant ACM Staff member, I want to have access to all approved versions of the curriculum recommendations.	
AS2	4e	As the relevant ACM Staff member, I want to be notified of posted changes to the curriculum.	

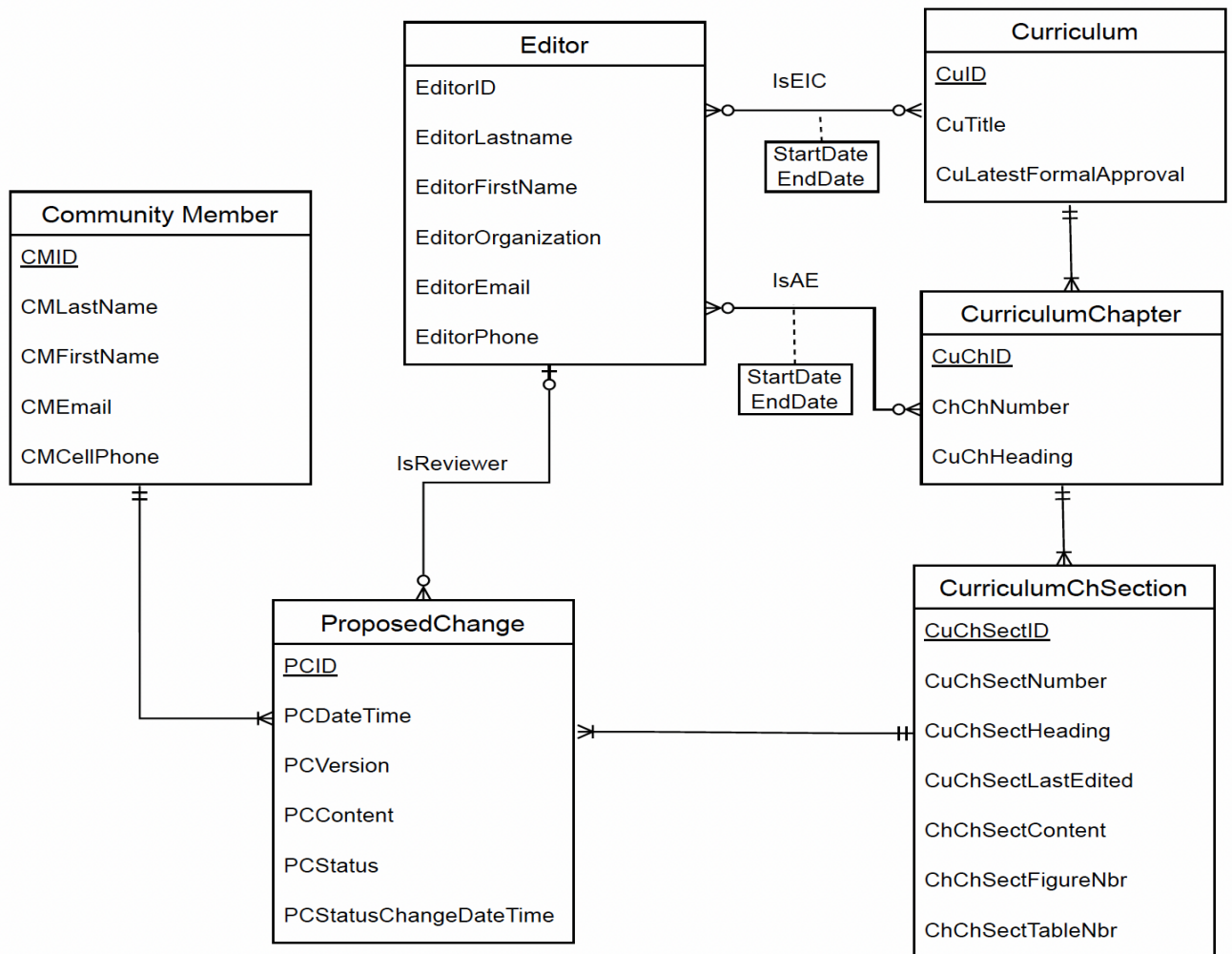
7. Data Requirements

- i. Logical data model E.g., entity-relationship diagrams and UML class diagrams. You may provide a data model for the business operations or

the data that the system modifies. Not the same thing as a database design data model.

- ii. Data dictionary Composition of data structures, meaning, data type, length, format, and allowed values.

Initial Conceptual Data Model



8. External Interface Requirements

- i. User Interfaces

Describe the logical characteristics of each interface between the software

product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed.

ii. **Hardware Interfaces**

Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.

iii. **Software Interfaces**

Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software components.

iv. **Communication Interfaces**

Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on.

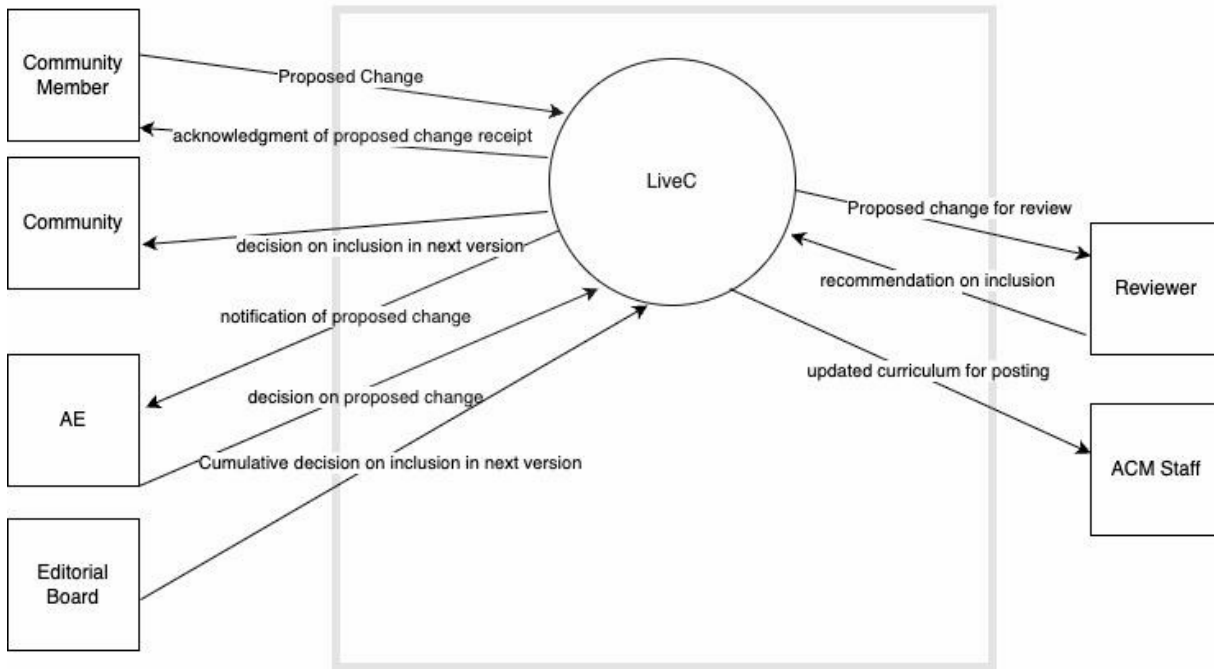
9. Software Quality Attributes

Specify requirements that include performance, security, reusability, maintainability, usability, availability, interoperability, etc.

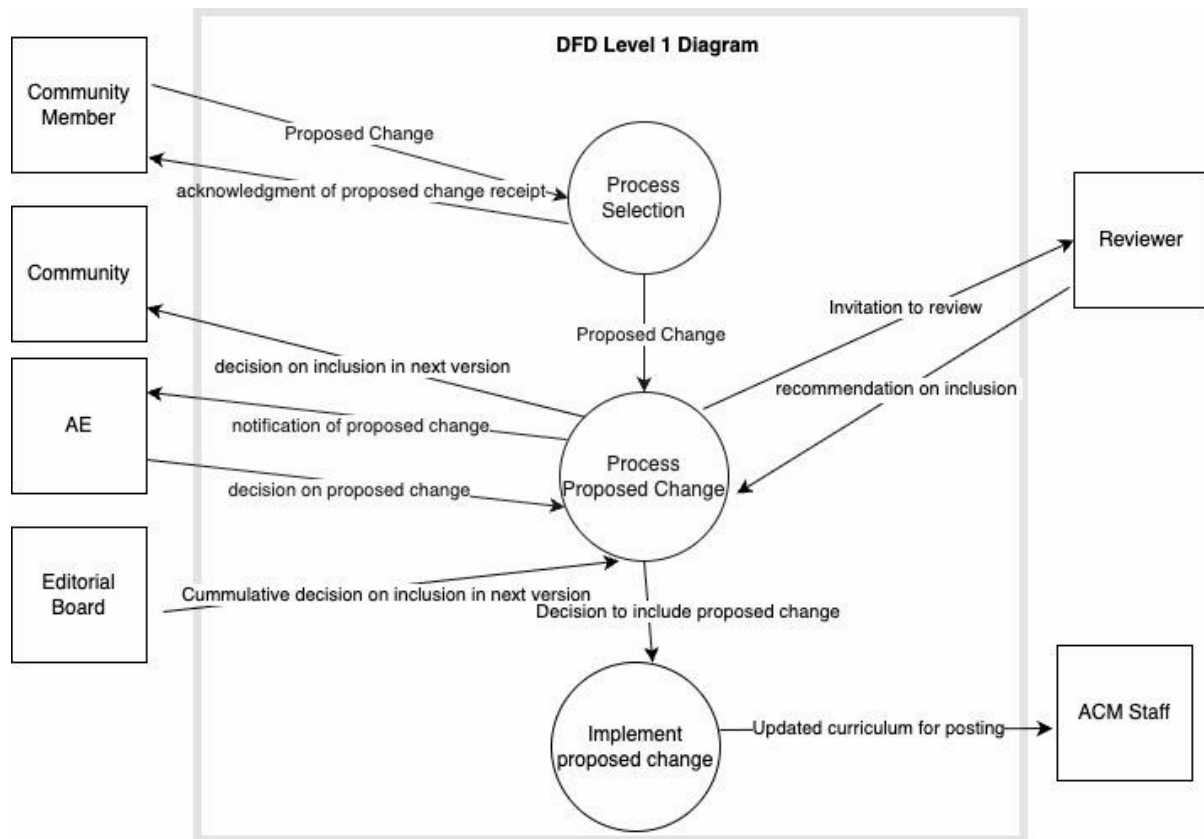
10. Analysis Models

Data Flow Diagrams

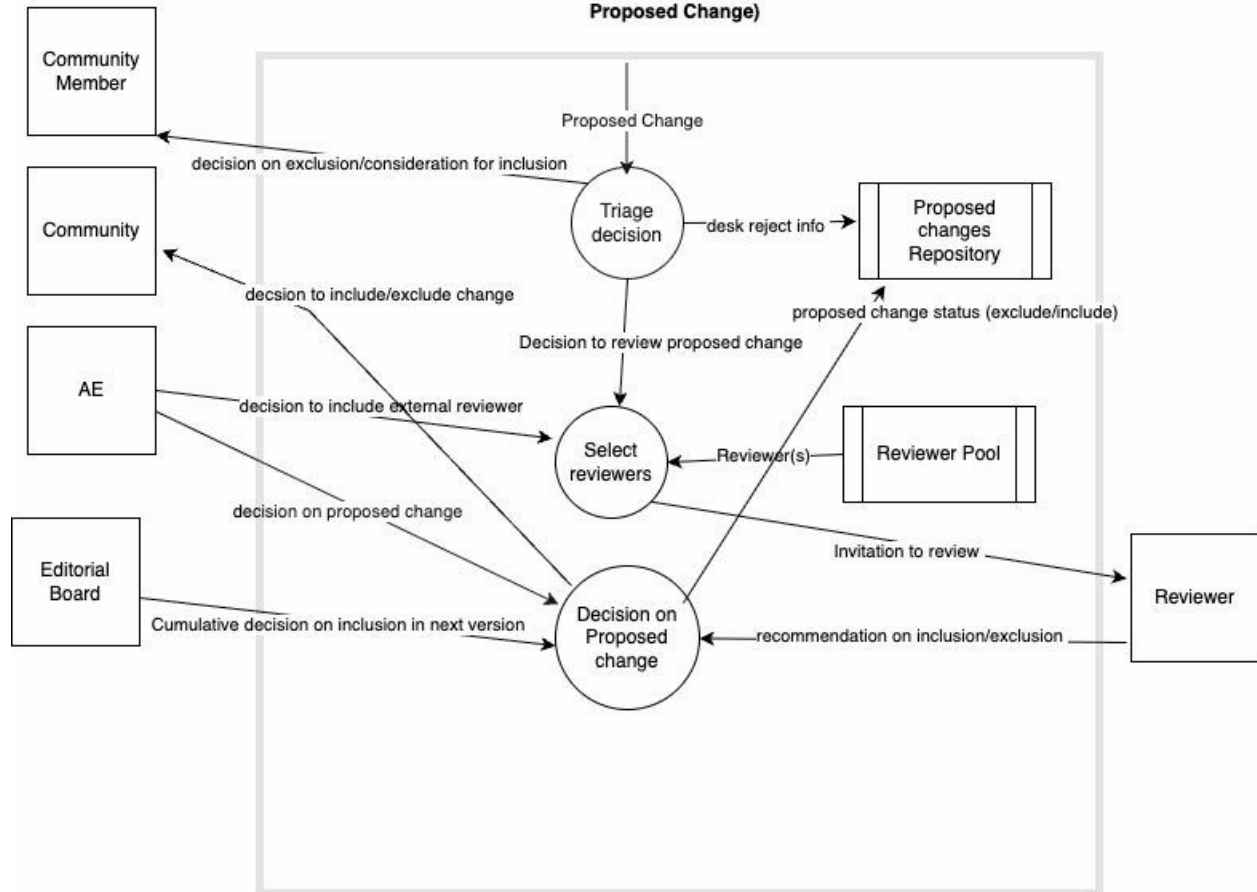
Context Diagram (DFD Level 0)



DFD Level 1 Diagram

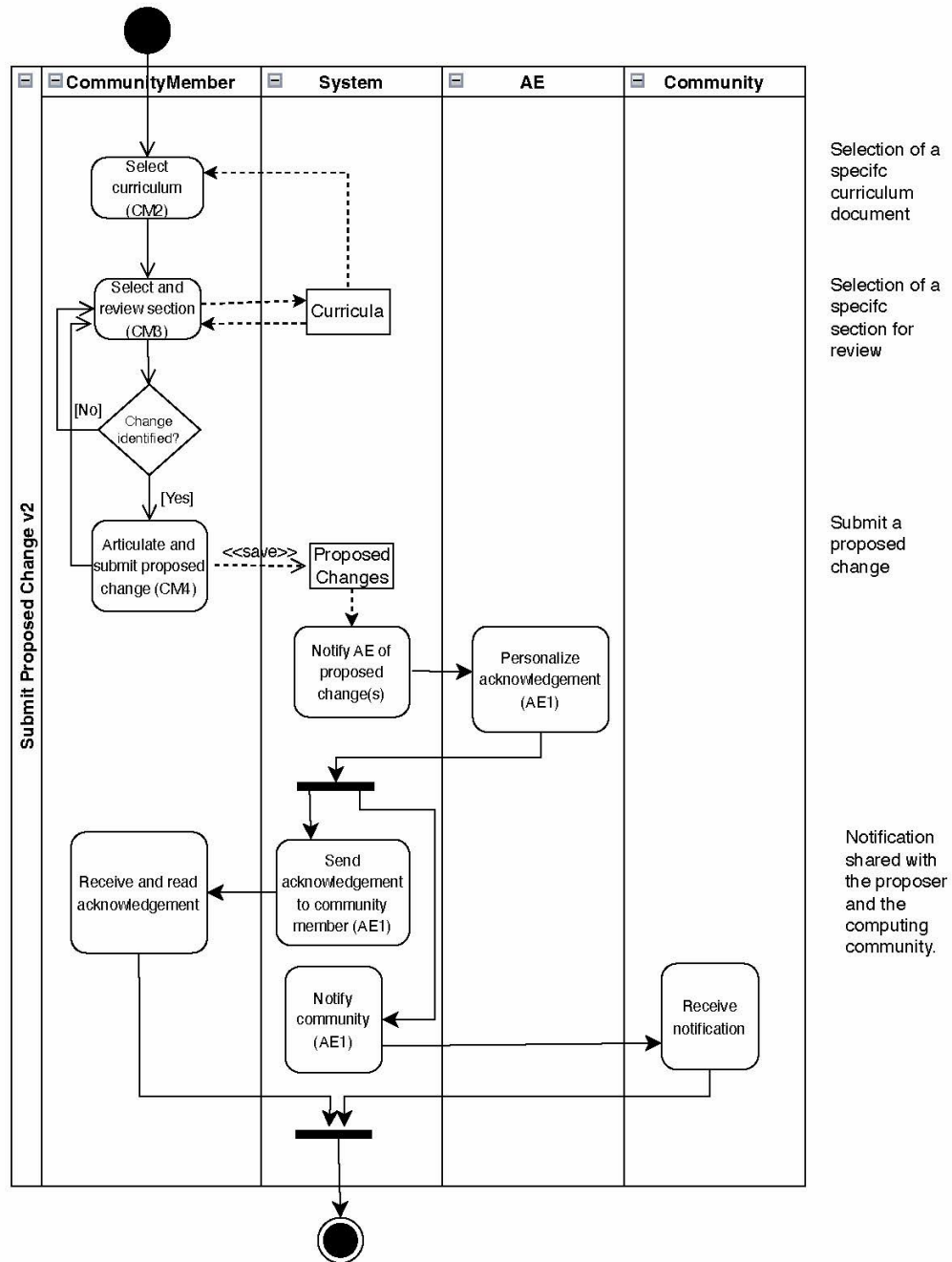


DFD Level 2 Diagram (Process
Proposed Change)

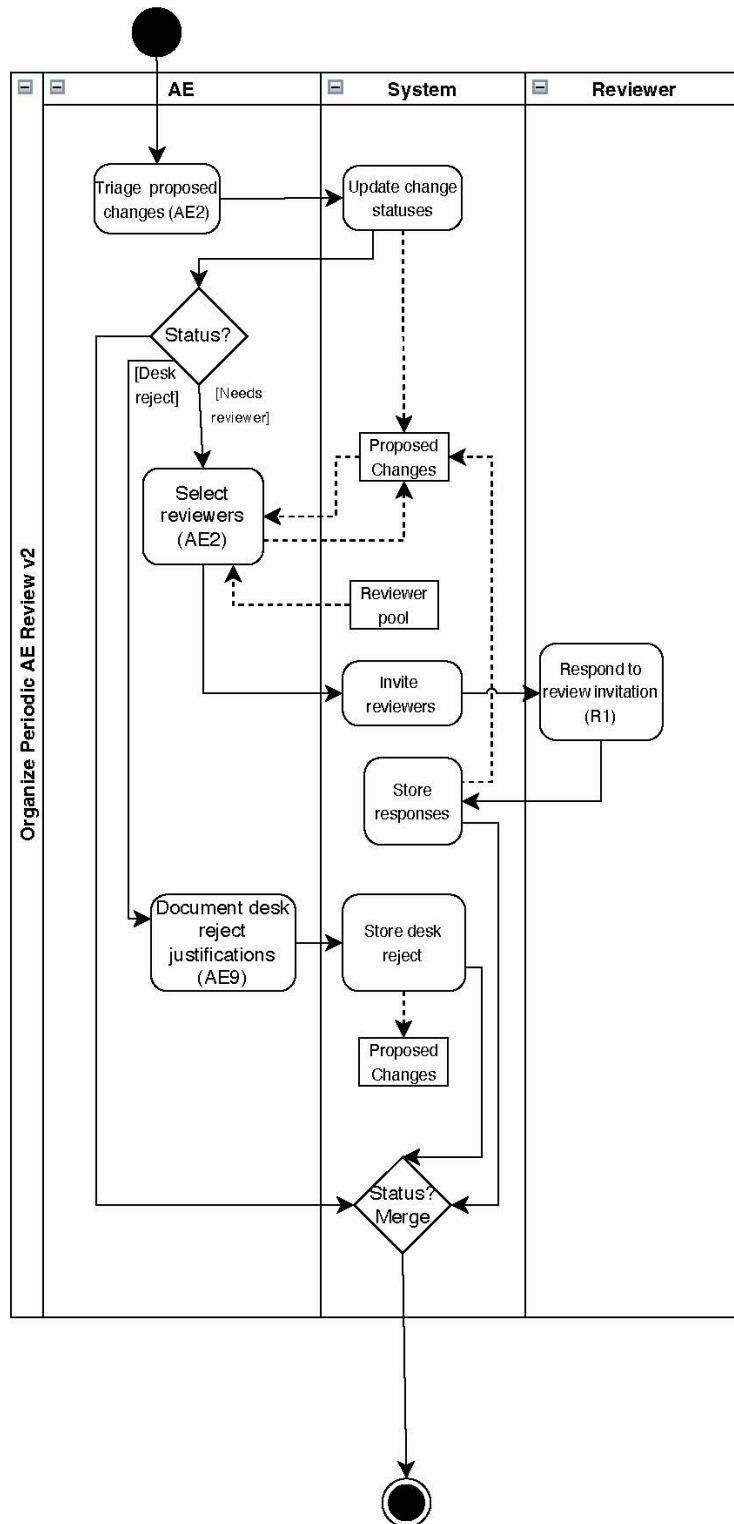


Submission of a proposed change by (computing) community members

Note: only registered users may propose changes.

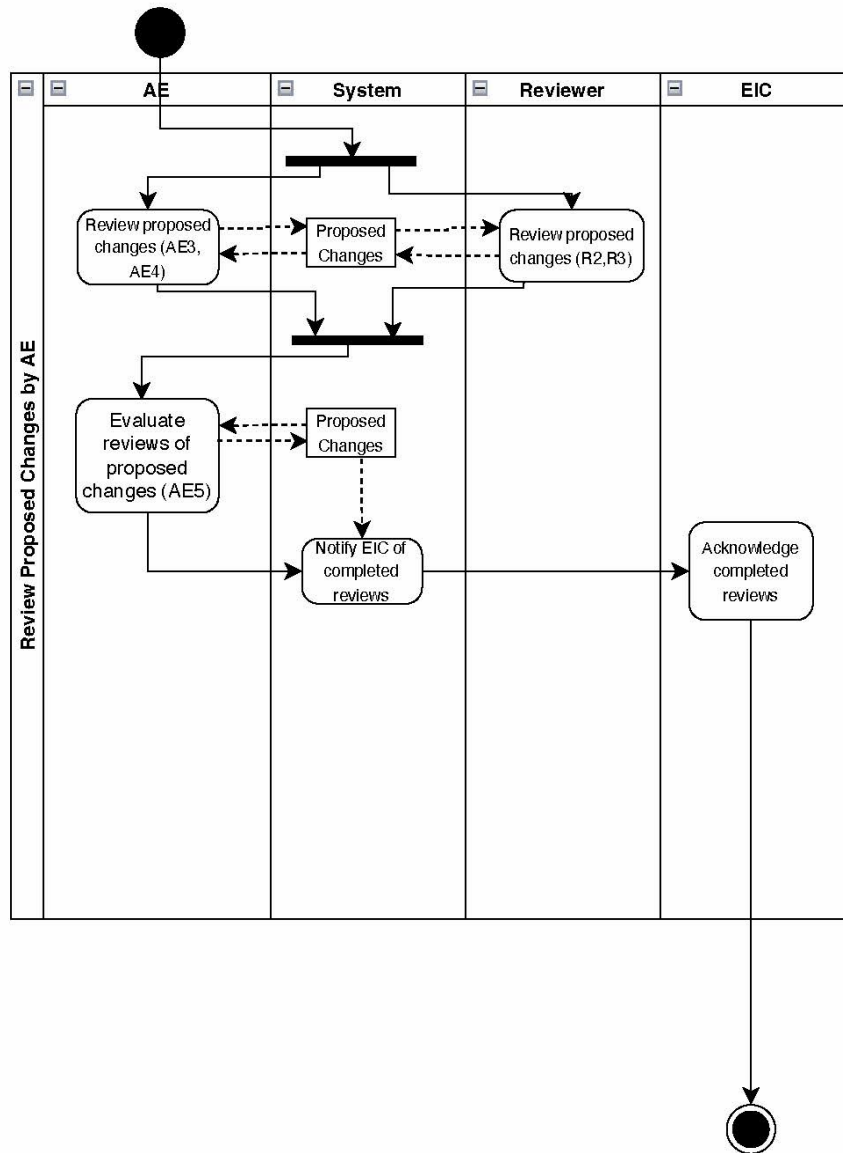


Organizing Proposed Changes for Review by the AE



This is a periodically repeated process that prepares a batch of proposals for review.

Reviewing of Proposed Changes by the AE and Reviewers



This is the actual proposed change review process that leads to recommendation(s) to the EIC.

Evaluating of the Proposed Changes by the EIC and Editorial Board

